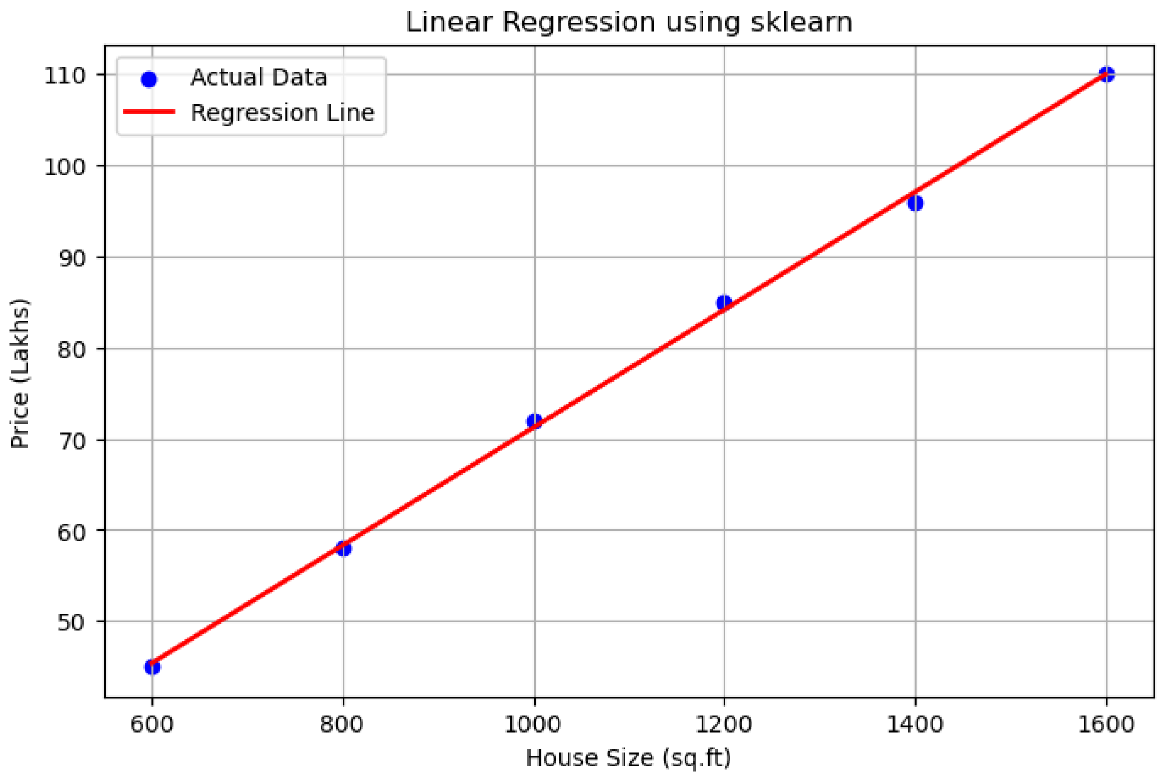
Assignment 3: Question 3, part 2

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Dataset
X = np.array([600, 800, 1000, 1200, 1400, 1600]).reshape(-1,1)
y = np.array([45,58,72,85,96,110])

# Create Model
model = LinearRegression()

# Train
model.fit(X,y)

# Parameters
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

# Prediction
y_pred = model.predict(X)

# Predict for 1500 sq.ft.
pred = model.predict([[1500]])
print("Predicted Price:", pred[0])

# Metrics
mse = mean_squared_error(y,y_pred)
r2 = r2_score(y,y_pred)

print("MSE:", mse)
print("R2 Score:", r2)

# Plot
plt.figure(figsize=(8,5))
plt.scatter(X,y,color='blue',label='Actual Data')
plt.plot(X,y_pred,color='red',linewidth=2,label='Regression Line')

plt.xlabel("House Size (sq.ft)")
plt.ylabel("Price (Lakhs)")
plt.title("Linear Regression using sklearn")
plt.legend()
plt.grid(True)

plt.show()
```

```
Slope: 0.06457142857142859
Intercept: 6.638095238095232
Predicted Price: 103.49523809523811
MSE: 0.4507936507936497
R2 Score: 0.9990741344461107
```


Linear Regression using sklearn
Actual Data
Regression Line
110
100
90
80
70
60
50
Price (Lakhs)
600
800
1000
1200
1400
1600
House Size (sq.ft)